

Script Title

Your Name

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1 Limits

Definition 1.1 (Limit). Let $f : D \rightarrow \mathbb{R}$ be a function with $D \subseteq \mathbb{R}$ and a a cluster point of D . Then L is the limit of f at the point a , if for all $\varepsilon > 0$ there exists a $\delta > 0$, such that for all $x \in D$ with $0 < |x - a| < \delta$ we have: $|f(x) - L| < \varepsilon$. We write:

$$\lim_{x \rightarrow a} f(x) = L.$$

2 Continuity

Definition 2.1 (Continuity). A function $f : D \rightarrow \mathbb{R}$ with $D \subseteq \mathbb{R}$ is continuous at the point $a \in D$, if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

A function is continuous if it is continuous at every point in its domain.

Bibliography